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CHM 413 — Past Questions & Materials
by Renance

Course: CHM 413
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 3
Levels: 400

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Paper 1 of 3 — CHM 413 Organometallic Chemistry and Industrial Applications

Level 400 • First Semester • PastQuestion

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Federal University of Petroleum Resources, Effurun
Department of Chemistry

B.Sc (Hons.) Degree Examination, First Semester 2024/2025 Session; May 2025
CHM 413- Organometallic Chemistry and Industrial Applications (2 credits)

Answer any **Three** questions only.

Time Allowed- 2 hrs.

1. (a) Describe the Dewar - Chatt - Duncanson model for bonding in metal carbonyls. (10 marks) ✓
 (b) How do the CO stretching frequencies in metal carbonyls relate to the metal-carbonyl bonding? (8 marks)
 (c) Draw the structure of $\text{Mn}_2(\text{CO})_{10}$, $\text{Fe}(\text{CO})_5$ and $\text{Fe}_2(\text{CO})_{12}$. (6 marks)
2. (a) Explain the role of π - backbonding in metal carbonyls. (5 marks)
 (b) Explain the relative position of the IR stretching absorptions in $[\text{V}(\text{CO})_6]^-$ versus $\text{Cr}(\text{CO})_6$. (5 marks)
 (c) What is the specific reaction used to prepare iron pentacarbonyl? (5 marks)
 (d) How does sigma- donation occur in metal carbonyls? (5 marks)
 (e) How is ferrocene synthesized? (4 marks).
3. (a) What are the advantages and disadvantages of using heterogeneous catalysts in organometallic chemistry? (3 marks) ✓
 (b) How can infrared spectroscopy be used to monitor the reaction of metal carbonyls with other molecules? (8 marks)
 (c) Write a balanced equation with an appropriate organometallic catalyst showing the overall (net) reaction in each of the following processes:
 i. Carbonylation of alkene to acyl products. (5 marks)
 ii. Oxidation of ethene to ethanal. (4 marks) ✓
 iii. Hydroformylation of olefin to aldehyde. (4 marks)
4. (a) What is hapticity in organometallic compounds? (3 marks) ✓
 (b) What is the difference between η_1 and η_5 hapticity? (6 marks) ✓
 (c) List the factors that influence the hapticity of a ligand in an organometallic Compounds. (5 marks)
 (d) Describe the structure of ferrocene. (4 marks) ✓
 (e) Briefly explain Friedel - Craft acylation of ferrocene. (6 marks) ✓

Paper 2 of 3 — CHM 413 ORGANOMETALLIC CHEMISTRY AND INDUSTRIAL APPLICATIONS 2024/2025

Level 400 • First Semester • PastQuestion

Federal University of Petroleum Resources, Effurun

Department of Chemistry

B.Sc (Hons.) Degree Examination, First Semester 2024/2025 Session; May 2025

CHM 413- Organometallic Chemistry and Industrial Applications (2 credits)

Answer any Three questions only. Time Allowed- 2 hrs.

1. (a) Describe the Dewar — Chatt - Duncanson model for bonding in metal carbonyls.

(10 marks)

(b) How do the CO stretching frequencies in metal carbonyls relate to the metal-carbonyl bonding? (5 marks)

(c) Draw the structure of $Mn_2(CO)_{10}$, $Fe(CO)_5$ and $Fe_3(CO)_9$. (6 marks)

2. (a) Explain the role of π -backbonding in metal carbonyls. (5 marks)

(b) Explain the relative position of the IR stretching absorptions in $[V(CO)_6]^-$ versus $Cr(CO)_3$. (5 marks)

(c) What is the specific reaction used to prepare iron pentacarbonyl? (5 marks)

(d) How does sigma-donation occur in metal carbonyls? (5 marks)

(e) How is ferrocene synthesized? (4 marks).

3. (a) What are the advantages and disadvantages of using heterogeneous catalysts in organometallic chemistry? (3 marks) | Per

(b) How can infrared spectroscopy be used to monitor the reaction of metal carbonyls with other molecules? (8 marks) |

(c) Write balanced equations with appropriate organometallic catalyst showing the overall (net) reaction in each of the following processes:

i. Carbonylation of alkene to aldehydes. (5 marks)

ii. Oxidation of ethene to ethanal. (4 marks)

iii. Hydroformylation of olefin to aldehyde. (4 marks)

4. (a) What is hapticity in organometallic compounds? (3 marks)

(b) What is the difference between η^1 and η^5 hapticity? (6 marks)

(c) List the factors that influence the hapticity of a ligand in an organometallic compound. (5 marks)

(d) Describe the structure of ferrocene. (4 marks)

(e) Briefly explain Friedel Craft acylation of ferrocene. (6 marks)

Paper 3 of 3 — CHM 413 ORGANOMETALLIC CHEMISTRY AND INDUSTRIAL APPLICATIONS 2024/2025

Level 400 • First Semester • PastQuestion

Federal University of Petroleum Resources, Effurun

Department of Chemistry and Science

B.Sc (Hons.) Degree Examination, First Semester 2024/2025 Session; awning

CHM 413- Organometallic Chemistry and Industrial Applications (2 credits)

Time allowed- 2 hrs.

Answer any Three questions only. Time Allowed

1. (a) Describe the Dewar Chatt - Duncanson model for bonding in metal carbonyls.

(10 marks) . _

(b) How do the CO stretching frequencies in metal carbonyls relate to the metal-carbonyl bonding? (6 marks)

(c) Draw the structure of $Mn_2(CO)_{10}$, $Fe(CO)_5$ and $Fe(CO)_2$. (6 marks)

2. (a) Explain the role of π -backbonding in metal carbonyls. (9 marks)

(b) Explain the relative position of the IR stretching absorptions in $V(CO)_6^{+}$ and $Cr(CO)_6$. (5 marks)

(c) What is the specific reaction used to prepare iron pentacarbonyl? (5 marks)

(d) How does sigma-donation occur in metal carbonyls? (5 marks)

(e) How is ferrocene synthesized? (4 marks).

3. versus

(a) What are the advantages and disadvantages of using heterogeneous catalysts in organometallic chemistry? (3 marks) a f

(b) How can infrared spectroscopy be used to monitor the reaction of metal carbonyls with other molecules? (8 marks)

(c) Write a balanced equation with an appropriate organometallic catalyst showing the overall (net) reaction in each of the following processes:

i. Carbonylation of alkene to acyl products. (5 marks)

ii. Oxidation of ethene to ethanal. (4 marks) Y

iii. Hydroformylation of olefin to aldehyde. (4 marks)

4. (a) What is hapticity in organometallic compounds? (3 marks) ¥

(b) What is the difference between η^1 and η^5 hapticity? (6 marks) v

(c) List the factors that influence the hapticity of a ligand in an organometallic compound. (5 marks)

(d) Describe the structure of ferrocene. (4 marks) = -¥

(e) Briefly explain Friedel-Crafts acylation of ferrocene. (6 marks)